



Bilingual children's Development of Cantonese: Responses to Cantonese A-not-A questions

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Cross-linguistic influence in bilingual acquisition

- Children acquire two languages as two largely independent systems
- Cross-linguistic influence (CLI): evidence for the effect of one language on the other (Serratrice, 2013)
- Hulk & Müller's (2000) CLI hypothesis:
 - Interface condition: CLI would take place between the discourse-pragmatic interface and different linguistic modules (e.g., syntax, semantics and phonology)
 - Overlap condition: Partial overlap in the surface levels of the two languages would be necessary for the occurrence of CLI
- Language dominance may determine the directionality of CLI (Yip & Matthews, 2000)
 - The more dominant in English, the more CLI from English
 - Evidence of transfer from a non-dominant language (e.g., English) to a dominant language (e.g., Cantonese) is only attested in some vulnerable domains

Cross-linguistic influence in bilingual acquisition

- Vulnerable domain
 - “A specific area of grammar prone to CLI regardless of patterns of dominance” (Yip & Matthews, 2007)
 - Usually created by a partial overlap where a language allows two possible structures (ambiguous input) but another language allows only one of the two
 - An example from Yip & Matthews (2007)
 - Cantonese allows different VPs to carry preverbal and postverbal PPs whilst English VPs allow only postverbal ones, making the placement of PPs a vulnerable domain
 - Non-target Cantonese postverbal PPs are attested in the production of Cantonese-English bilingual children

English yes/no questions

- Invariant response particles
 - “Yes” to show affirmation
 - “No” to show negation
- Assign polarity value to the proposition as in (2) (Holmberg, 2016)

(1) Q: Do you want tea?

A: Yes

(2) [_{CP} Do [_{IP} you [_{±Pol}] want tea]]

[_{FocP} Yes [_{IP} I [+Pol] want tea]]

- The IP (or VP) later undergoes ellipsis

Cantonese yes/no questions

- Particle and intonational questions
- Invariant response particle
 - *Hai6* to show affirmation
 - *m4hai6* to show negation

(3) Q: Nei5 haa6 go3 lai5baai3 fong3 gaa3 aa4 ?

you next CL week take leave SFP

“You’re going on leave next week?”

(Matthews & Yip, 2011)

A: Hai6 aa3

Be SFP

“Yes.”

- Propositional anaphors that pick up a discourse referent to assert or negate it
- Hypothesized to be located at ActP in the CP domain (Krifka, 2013; Cheng, 2022)
- *Hai6/m4hai6* is the assertion/negation of the discourse referent (i.e., the IP) on its own without any further syntactic structures (Cheng, 2022), as in (4)

(4) $\llbracket [\text{ActP yes}] \rrbracket = \text{ASSERT}(d)$, where d is the propositional discourse referent

Cantonese yes/no questions

- *Hai6/m4hai6*
 - **Can** be answer to particle and intonational questions (and declarative sentences) as they contain only one discourse referent each time for *hai6/m4hai6* to pick up
 - **Cannot** be answer to neutral polar questions (i.e., A-not-A questions) as it contains two referents (+ and -) to pick up
- This anaphora-based analysis explains why it cannot be used to answer A-not-A questions
- If *hai6/m4hai6* has an ellipsis-based structure like “yes”/“no” does, it should be able to answer A-not-A questions as “yes”/“no” should (which it cannot)

(5) Q: Gin6 saam1 leng3-m4-leng3 aa3 ?

CL dress pretty-not-pretty SFP

“Does the dress look good?”

(Matthews & Yip, 2011)

A: *Hai6 aa3

Be SFP

* “Yes, it does.”

Cantonese yes/no question

- A-not-A questions

(6) Q: Gin6 saam1 leng3-m4-leng3 aa3 ?

CL dress pretty-not-pretty SFP

“Does the dress look good?”

(Matthews & Yip, 2011)

A: Leng3 aa3

Pretty SFP

“Yes, it does.”

- Verb echo answer (VEA)
 - Reduplicating the verb or adjective
 - “A” to show affirmation, “not A” to show denial
- The syntactic derivation of VEA varies across studies, but they share a common structure as (7) (e.g., Holmberg (2016), Wei (2019), Chen (2022) etc.)

(7) [_{CP} V_i [_{TP} ... t_i ...]]

Cross-linguistic influence

- Both Interface and Overlap conditions predict CLI to occur
- Interface Condition
 - Response particles are situated at the syntax-discourse interface
- Overlap Condition
 - Cantonese yes/no questions allow both invariant responses (i.e., *hai6* and *m4hai6*) and variable responses (i.e., VEA)
 - English yes/no questions can only be answered with invariant responses (i.e., “yes” and “no”)
 - Such partial structural overlap makes the response to Cantonese yes/no questions a vulnerable domain

Cross-linguistic influence

- Input ambiguity
 - When a bilingual child hears a Cantonese particle/intonational question answered by *hai6/m4hai6*, this input is consistent with both anaphora-based and the ellipsis-based analyses
 - Favours the ellipsis-based analysis as it is the only structure consistent with the distribution of “yes”/“no” in English
 - Bilingual children’s misanalysis on the structure of *hai6/m4hai6* may result in using *hai6/m4hai6* as response particles to all Cantonese yes/no questions

Anaphora-based analysis
[[[ActP *Hai6*]] = ASSERT(d)



Hai6 as a response to
Cantonese yes/no question

Ellipsis-based analysis
[_{FocP} *Hai6* [_{IP} ...]]



Ellipsis-based analysis
[_{FocP} *Yes* [_{IP} ...]]



“Yes” as a response to
English yes/no question

Cross-linguistic influence

- However, apart from English influence, it may also result from overgeneralization of using *hai6/m4hai6* as response particles (within Cantonese)
 - If so, non-target *hai6/m4hai6* responses should also occur in monolingual children
 - Interaction of this overgeneralization with CLI will result in a quantitative difference in the production of non-target *hai6/m4hai6* responses (i.e., *hai6* mismatch) to A-not-A questions between bilingual and monolingual children

Research questions

1. To what extent do monolingual and bilingual children produce target responses and mismatches to A-not-A questions?
2. If they produce mismatches, what types of mismatches are produced?
3. Does language dominance profile of bilingual children affect their production rate of mismatch?

Hypotheses and Predictions

1. Overgeneralization of *hai6/m4hai6* as response particle will occur so *hai6* mismatches will occur in both bilingual and monolingual children.
2. CLI will occur in bilingual children's A-not-A responses, interacting with the overgeneralization of *hai6/m4hai6*. Therefore, bilingual children will produce more *hai6* mismatches than monolingual children.
3. Within bilingual children, the more Cantonese-dominant the child is, the less CLI from English there will be, thus the fewer *hai6* mismatches will be produced.

Method

- Corpora

1. Hong Kong Bilingual Child Language Corpus ($N = 8$, Yip & Matthews, 2007)

2. Hong Kong Cantonese Child Language Corpus ($N = 8$, Lee et al., 1996)

- The corpus data are analyzed by sampling children's utterances at 3-month intervals at ages 2;0, 2;3, 2;6, 2;9 and 3;0
- Only Cantonese data are considered
- Note that children from Cancorp are not strictly monolinguals. Some children have some limited exposure to English (e.g., Bernard was born in the United Kingdom and was brought back to Hong Kong at the age of 0;8)
- We calculated the number of A-not-A questions addressed to the target children, number of ignored questions, number of target responses, and number of mismatches

Hai6 mismatch

- Using copular verb *hai6* “yes” or *m4hai6* “no” to answer non-copular A-not-A questions

(8) LIN: Gam2 nei5 zung1-m4-zung1ji3 tai2 aa3 ?

So you like-not-like watch SFP

“So do you like to watch it?”

CHI: Hai6 aa3

Be SFP

“Yes.”

(Llywelyn, 2;06.20)

- The bilingual child Llywelyn responded with *hai6* ‘be’ instead of the target VEA *zung1ji3* ‘like’.

Jau5 mismatch

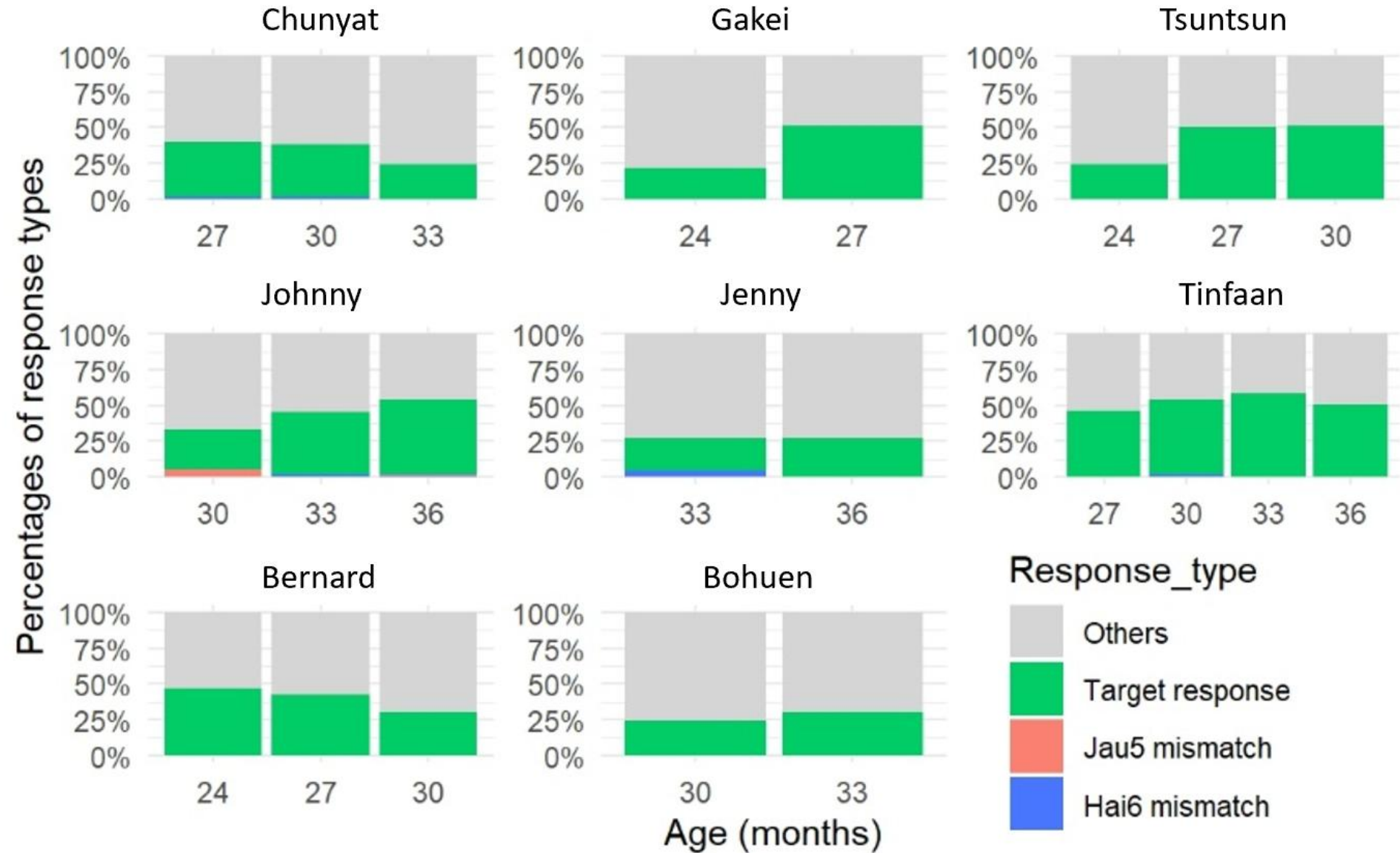
- Using *jau5* “have” or *mou5* “not have” to answer non-*jau5-mou5* A-not-A questions

(9) INV: Sik1-m4-sik1 mit1 ?
know-not-know peel ?
“Do you know how to peel it off?”
CHI: Jau5 aa3
Have SFP
“Yes.”

(Chunyat, 02;03.07)

- The monolingual child Chunyat responded with *jau5* ‘have’ instead of the target VEA *sik1* ‘know’

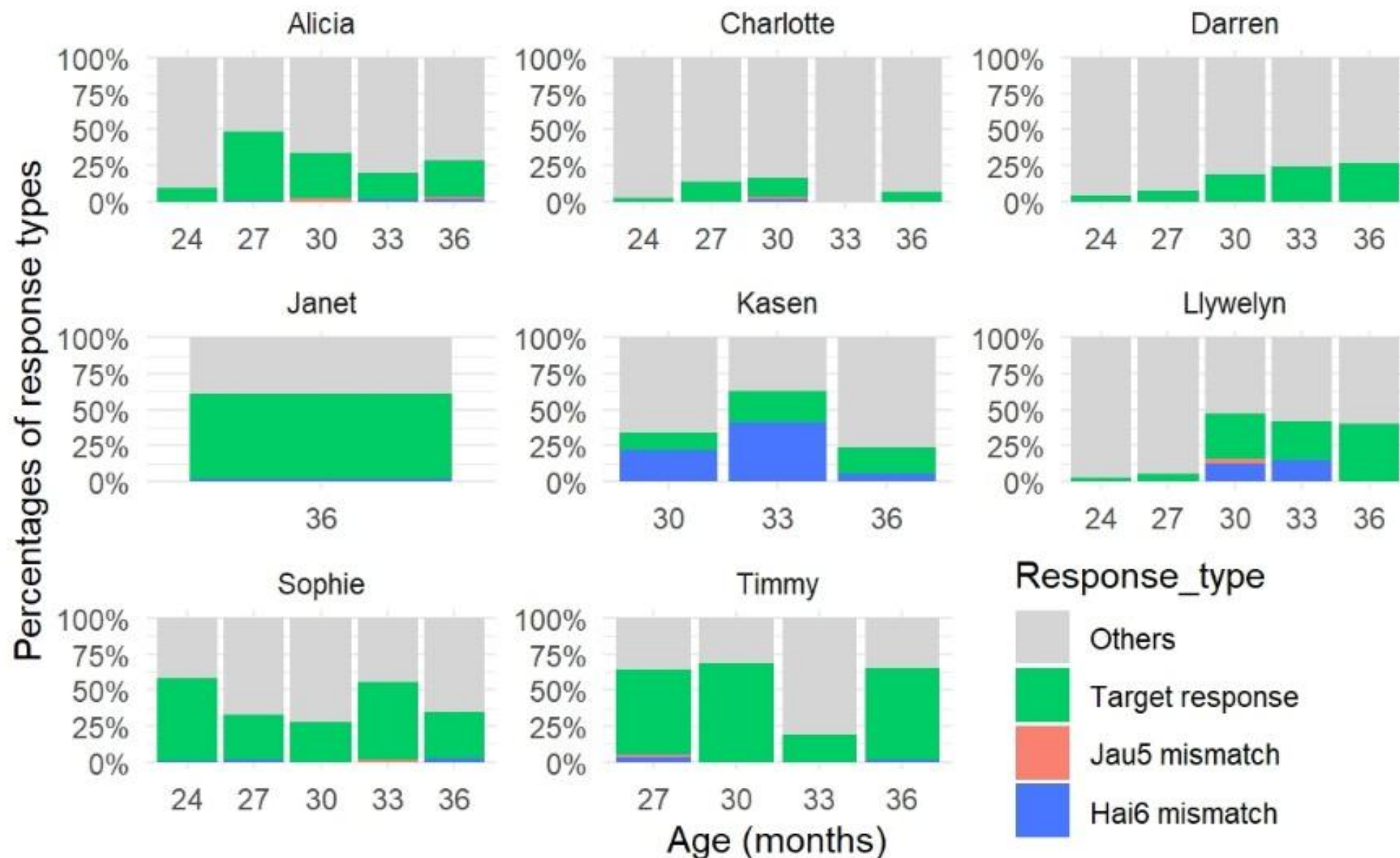
Monolingual Children's Response Types to A-not-A Questions



Mismatch production by monolingual children

- Only four children (Chunyat, Jenny, Johnny, and Tinfan) produced mismatches
 - All four produced *hai6* mismatch
 - Their production of each type of mismatch occurs at relatively low frequencies (generally < 5%)
- *Hai6* mismatch productions are attested in children who have some exposure to English
 - Jenny's domestic helper would speak some English to her
 - Tinfan would communicate in “something English-like” with the domestic helper
 - Chunyat's mother was a secondary school English teacher
 - Johnny was taken care by a Thai domestic helper

Bilingual Children's Response Types to A-not-A Questions

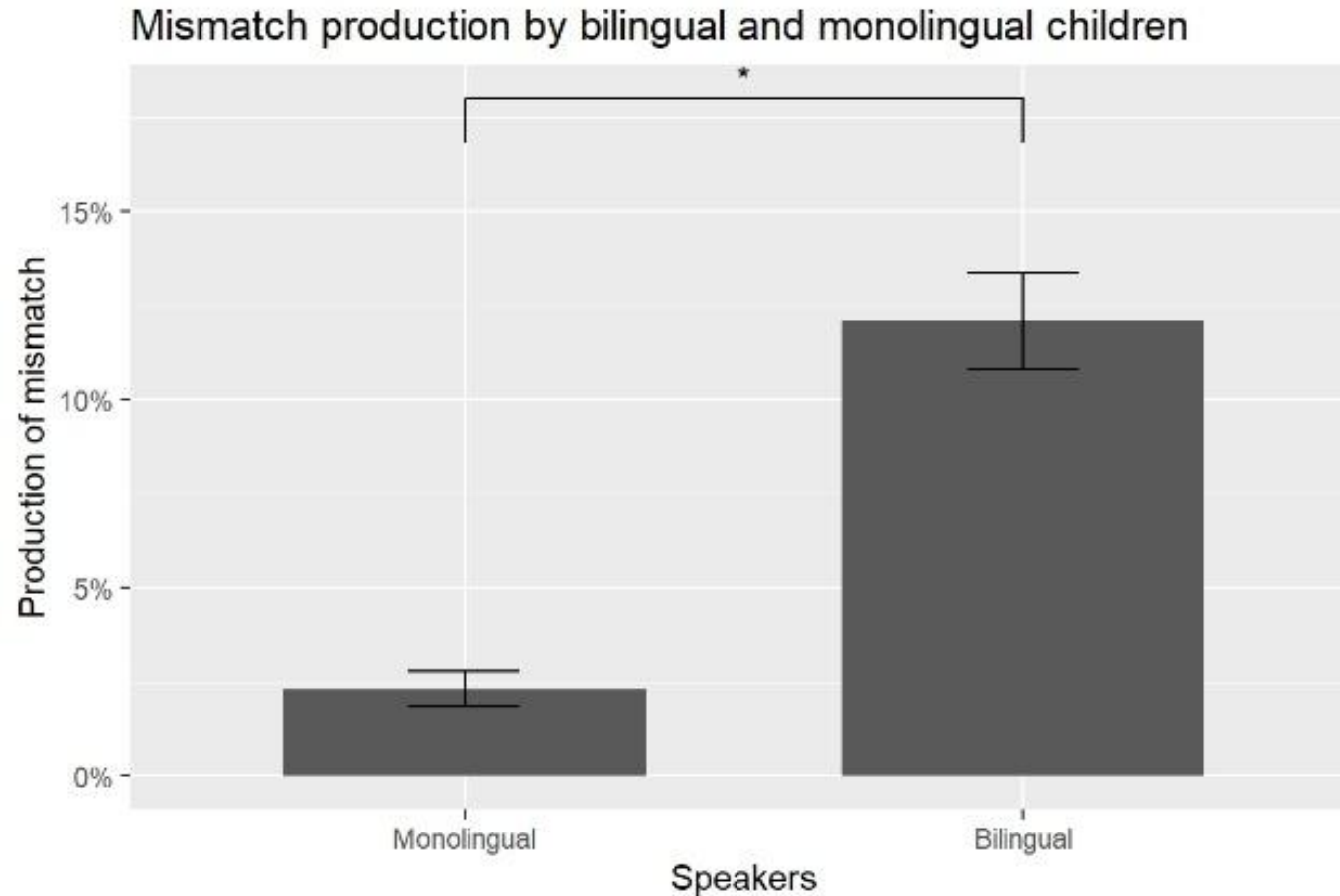


Mismatch production by bilingual children

- The production of type 1 mismatch was attested most frequently in the balanced bilingual child, Kasen
 - Nearly 40% at his peak 2;9
 - Cantonese-dominant children in general (except Llywelyn) produced type 1 mismatch at much lower frequencies ($< 10\%$)
 - English-dominant child Charlotte did not produce many mismatches
 - She frequently responded in English to Cantonese A-not-A questions
 - English responses were not counted as mismatch
- No mismatches were attested in Darren's production
 - He only responded to 3 A-not-A questions in his 2;3 recording and 4 in his 3;0 recording

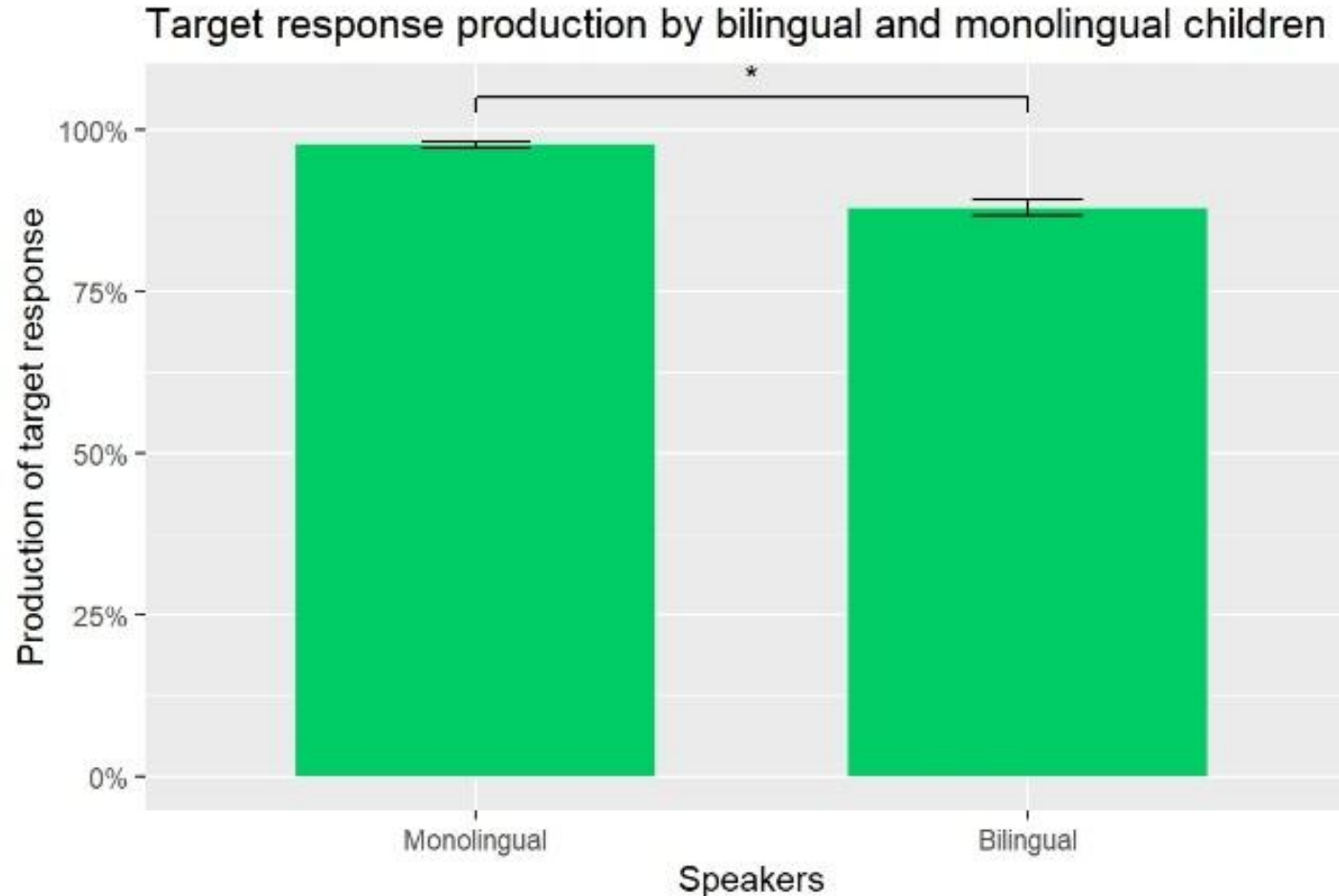
Mismatches

- Bilingual children overall produced significantly more mismatches than monolingual children ($\beta = 2.168$, $SE = .854$, $t(1658) = 2.540$, $p = .01$)



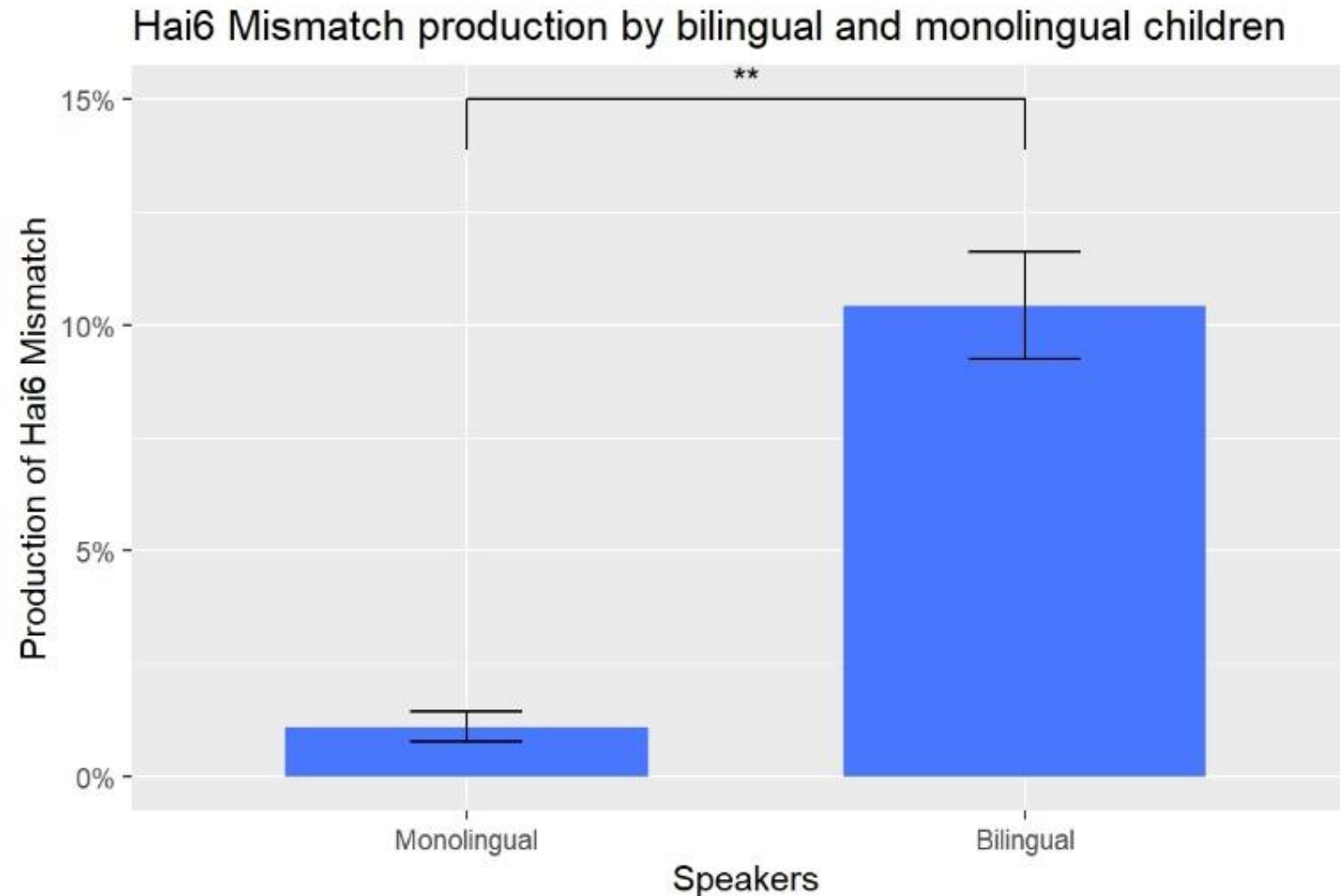
Target response

- Bilingual children overall produced fewer target responses than monolingual children ($\beta = -2.168$, $SE = .853$, $t(1658) = -2.540$, $p = .01$)



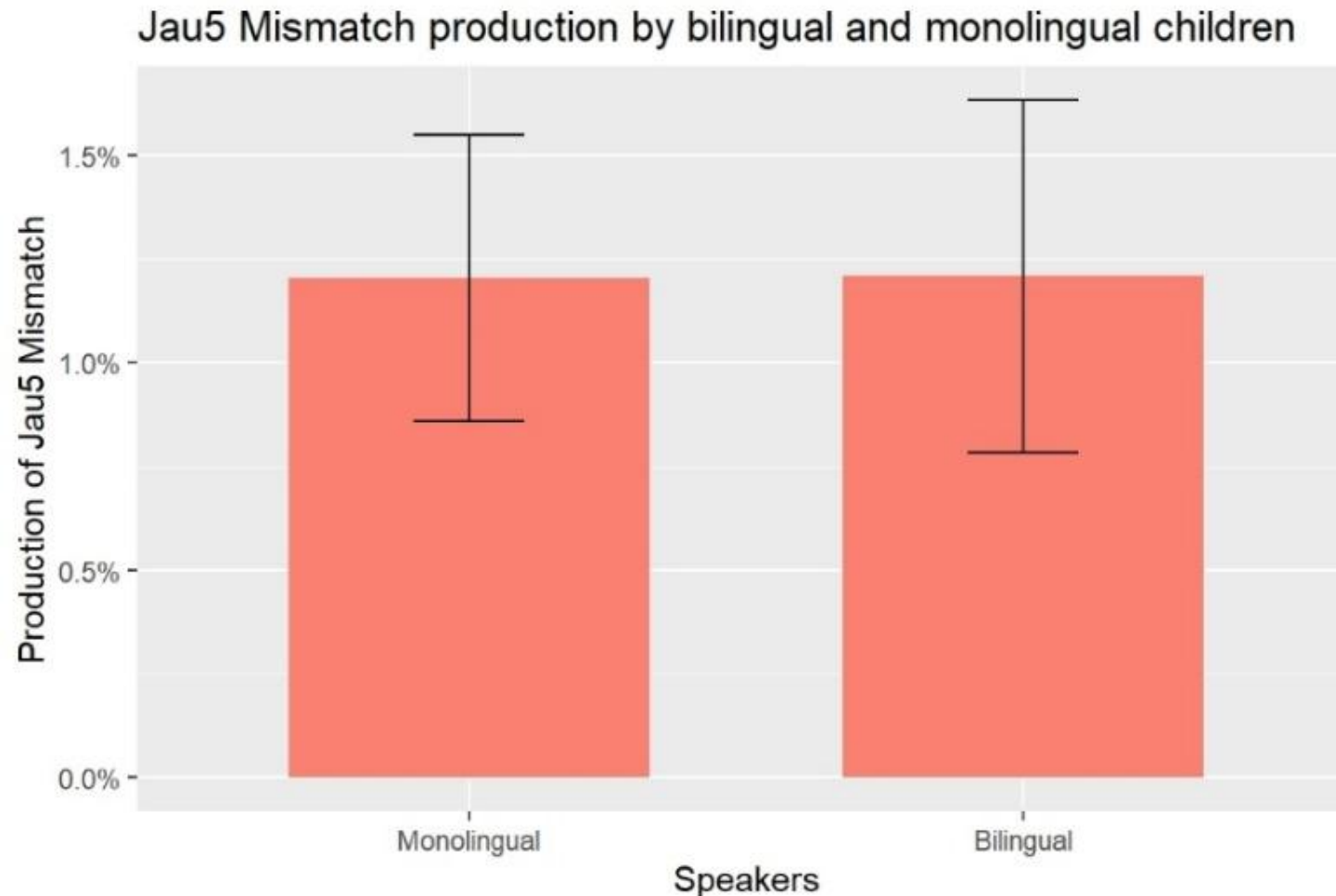
Hai6 mismatch

- Bilingual children produced significantly more *hai6* mismatch than monolingual children ($\beta = 2.232$, $SE = .844$, $t(1658) = 2.645$, $p = .008$)



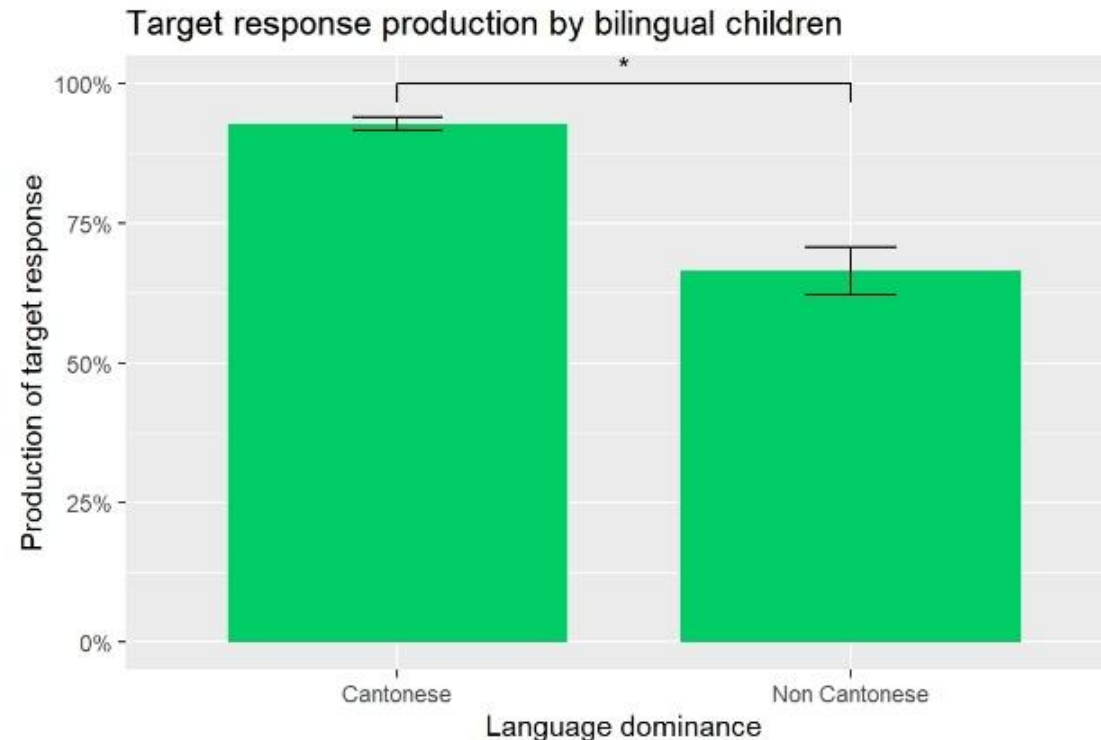
Jau5 mismatch

- There is no significant difference in the production of *jau5* mismatch between bilingual and monolingual children ($\beta = .999$, $SE = 1.06$, $t(1658) = .944$, $p = .345$)



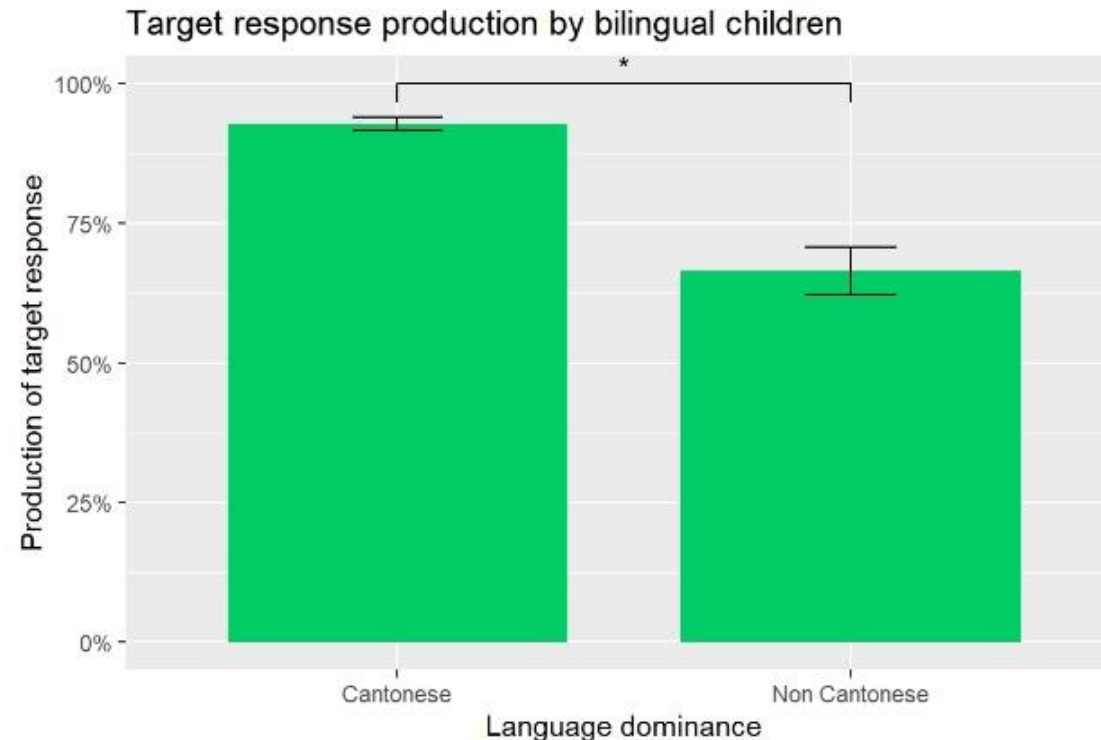
Language dominance as a factor

- Within bilingual children, we grouped them into Cantonese and Non-Cantonese dominant children
- Cantonese-dominant children produced more target response than non-Cantonese-dominant children ($\beta = -4.663$, $SE = 1.239$, $t(1948) = -3.763$, $p < .001$)



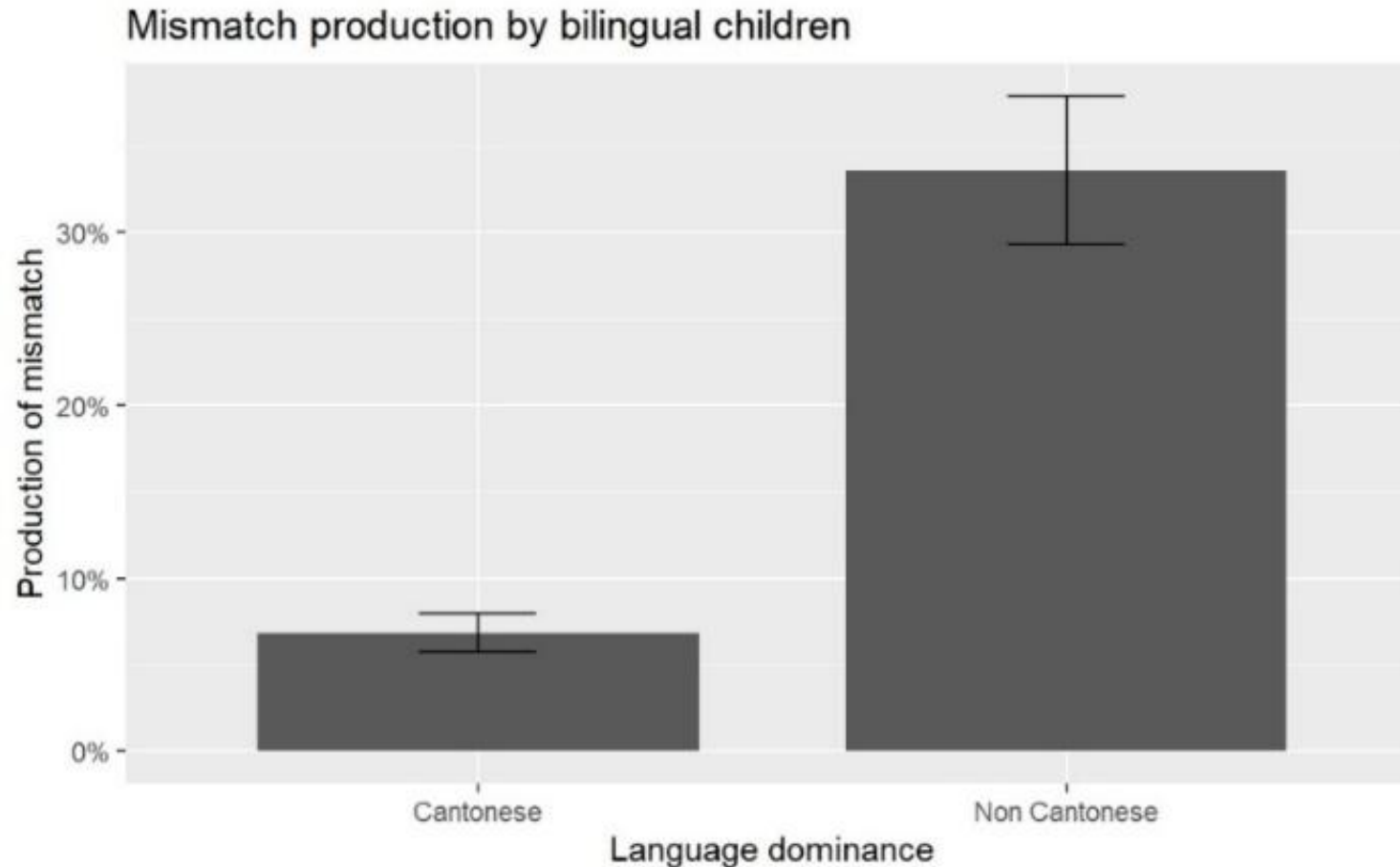
Target response within bilingual children

- Within bilingual children, we grouped them into Cantonese and Non-Cantonese dominant children
- Cantonese-dominant children produced more target response than non-Cantonese-dominant children ($\beta = -4.663$, $SE = 1.239$, $t(1948) = -3.763$, $p < .001$)



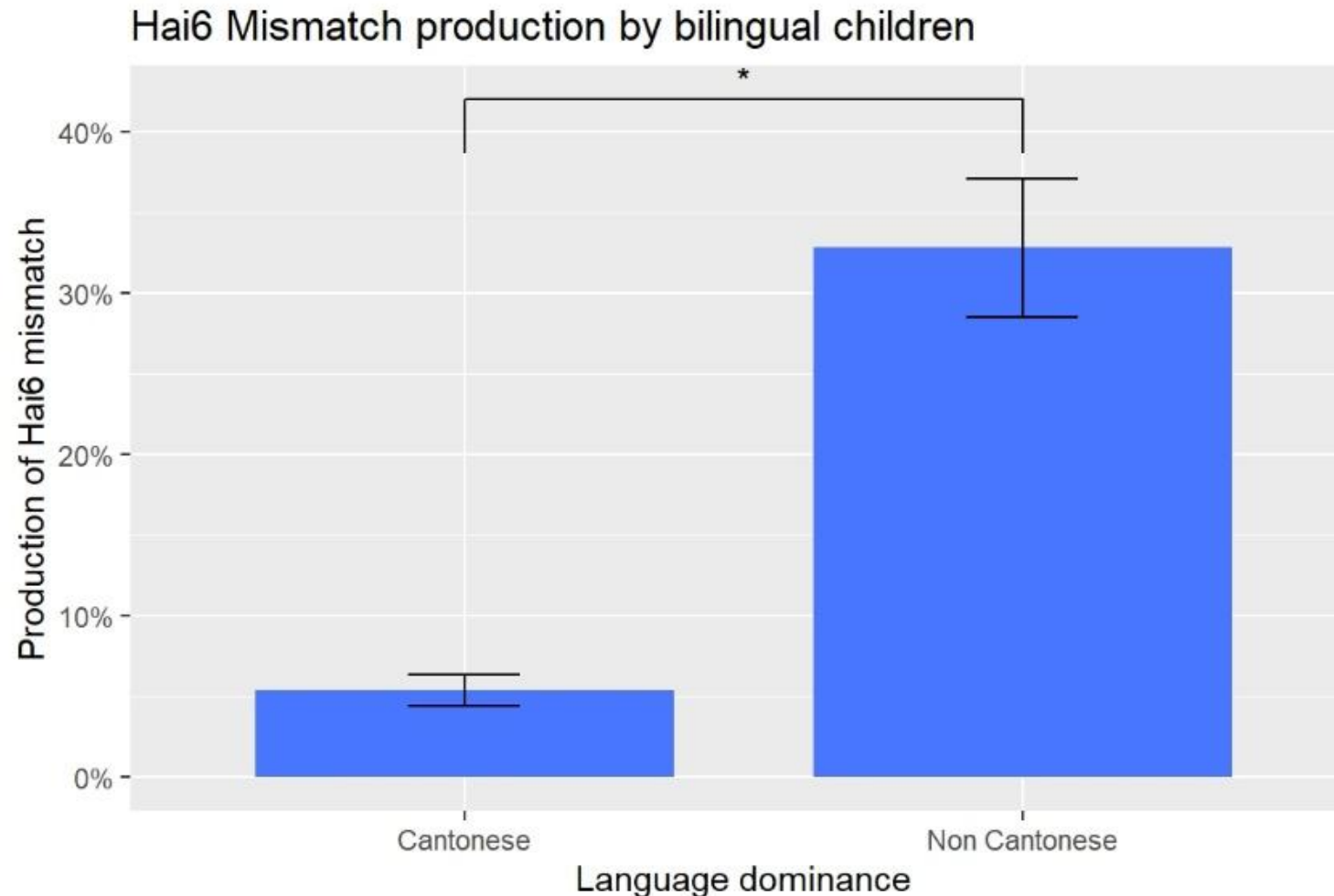
Mismatches within bilingual children

- No significant difference was found in the production of mismatches between Cantonese-dominant and non-Cantonese-dominant children ($\beta = 5.856$, $SE = 3.477$, $t(1948) = 1.684$, $p = .09$)



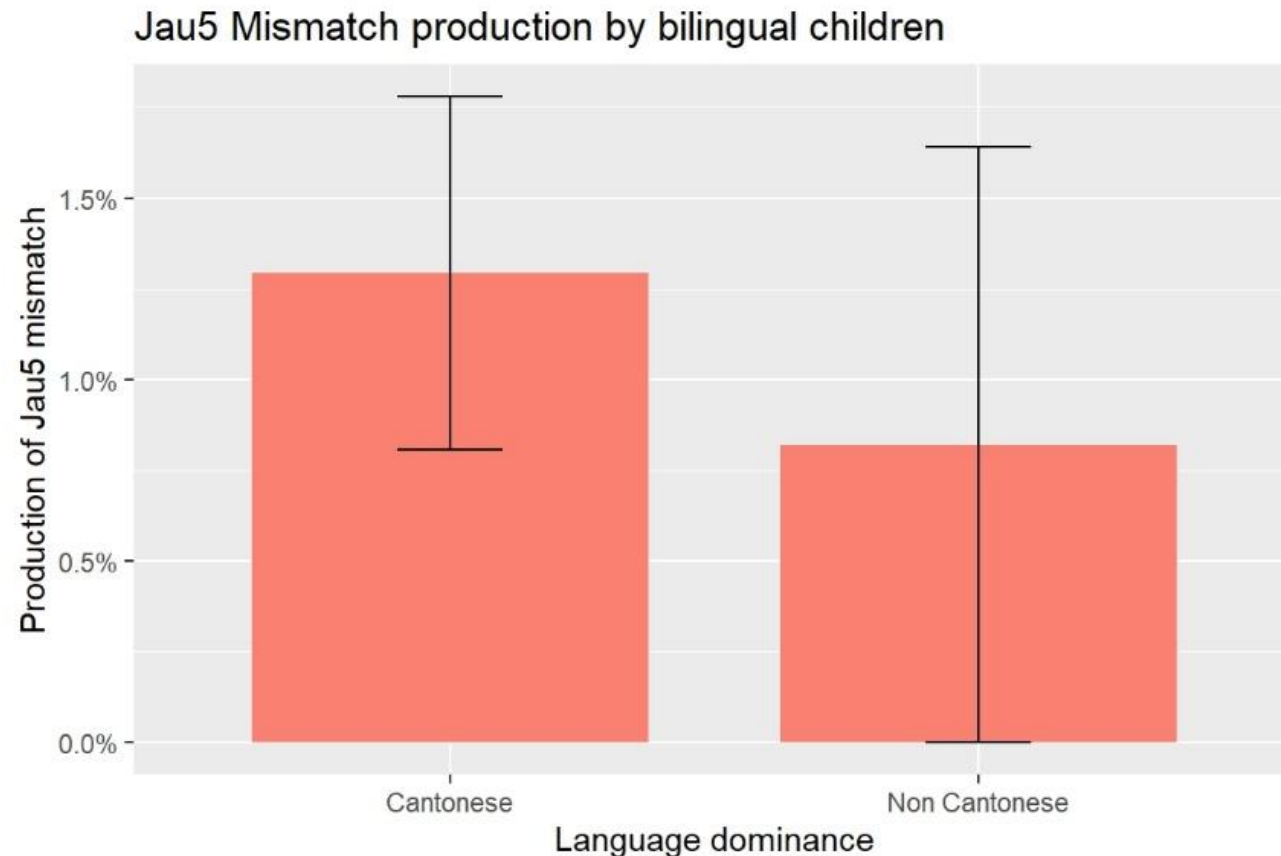
Hai6 mismatches within bilingual children

- Non-Cantonese-dominant children produced significantly more *hai6* mismatch than Cantonese-dominant children ($\beta = 5.969$, $SE = 2.839$, $t(1948) = 2.102$, $p = .04$)



Jau5 mismatches within bilingual children

- No significant difference is found in the production of *jau5* mismatch between Cantonese-dominant and non-Cantonese-dominant children ($\beta = 0.602$, $SE = 7.936$, $t(1948) = 0.076$, $p = .94$).



Discussion

- *Hai6* mismatches attested in the production of all bilingual children (except Darren)
 - Provides evidence for our prediction
 - CLI from a weaker language (English) to a dominant language (Cantonese) occurs in children's responses to Cantonese yes/no questions (vulnerable domain)
- Given the ambiguous input, bilingual children may apply the ellipsis-based analysis to *hai6/m4hai6*, the same as “yes”/“no” in English, so they used it to answer A-not-A questions
 - [_{FOCP} *Hai6* [_{HP} ...]]
- For those monolingual children who did not produce *hai6* mismatches, *hai6/m4hai6* has a structure (anaphora-based analysis) that prevents it from being used as an answer to neutral polar questions (i.e., A-not-A questions)

Discussion

- Quantitative difference in *hai6* mismatches between monolingual and bilingual children is consistent with our hypotheses 1 and 2:
 - Hai6 mismatch results from overgeneralization in monolingual children's production;
 - Hai6 mismatch results from interaction between overgeneralization and CLI in bilingual children's production
- However, *hai6* mismatches are found in the production of “monolingual” children with some (potential) English exposure
 - Overgeneralization of *hai6/m4hai6* as response particles was not attested in monolingual children with no known English exposure (i.e., Bernard, Bohuen, Gakei, and Tsuntsun)
 - *Hai6* mismatch might simply be a result of CLI

Discussion

- Cantonese-dominant children produced more *hai6* mismatches than non-Cantonese-dominant children
 - Language dominance plays a role, suggesting our second hypothesis to be correct
 - The most English-dominant child Charlotte did not produce the most *hai6* mismatches
- The unexpected *jau5* mismatch is not a result of CLI
 - No quantitative difference between bilingual and monolingual children
 - *Jau5* is frequently used as an answer to Cantonese yes/no questions
 - Might be a result of overgeneralization

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Thank you!

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